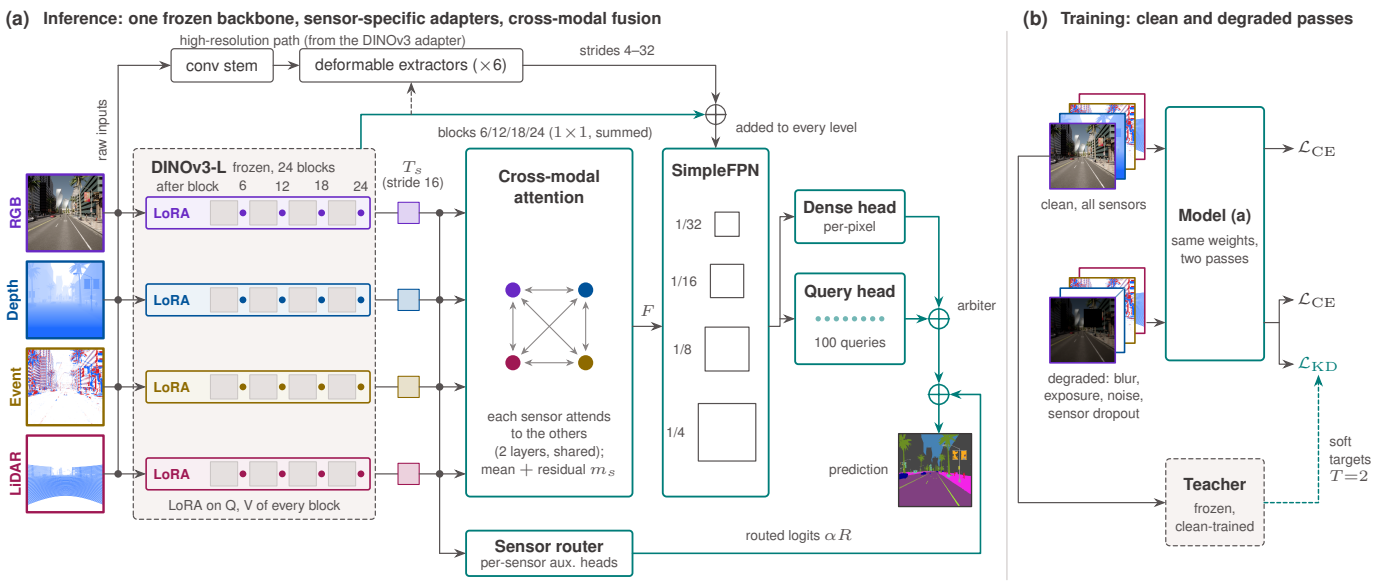

(a) Inference: one frozen backbone, sensor-specific adapters, cross-modal fusion
high-resolution path (from the DINOv3 adapter)
conv stem
deformable extractors (×6)
strides 4–32
raw inputs
blocks 6/12/18/24 (1 × 1, summed)
added to every level
DINOv3-L frozen, 24 blocks
after block 6 12 18 24
$T_s$ (stride 16)
RGB
Depth
Event
LiDAR
LoRA
LoRA
LoRA
LoRA
LoRA on Q, V of every block
Cross-modal attention
each sensor attends to the others (2 layers, shared); mean + residual $m_s$
$F$
SimpleFPN
1/32
1/16
1/8
1/4
Dense head
per-pixel
Query head
100 queries
arbiter
prediction
Sensor router
per-sensor aux. heads
routed logits $\alpha R$
(b) Training: clean and degraded passes
clean, all sensors
degraded: blur, exposure, noise, sensor dropout
Model (a)
same weights, two passes
$\mathcal{L}_{\mathrm{CE}}$
$\mathcal{L}_{\mathrm{CE}}$
$\mathcal{L}_{\mathrm{KD}}$
soft targets $T=2$
Teacher
frozen, clean-trained